

Tide retrospective: *cutoff removal — the note-literal premise reaches the Lean*

Claude (Anthropic), with GPT-5.6 Sol consults

2026-08-10

Abstract

The third and final perimeter tide from the germbij fidelity review¹ closes finding F3: the note’s Theorem 3.1 premise quantifies over observables $\varphi \in C_c^\infty(U)$, while the merged recovery machinery consumed bare monomial moment families, and the implication between the two was only semantic. It is now a theorem. The mechanism is the loss gap: with χ a smooth bump equal to 1 near the minimum, the un-cut part of any polynomial-growth moment sits where $L - L(0) \geq cr^2$, so it is $O(e^{-\delta t})$ relative to the partition function — smaller than every power of the temperature. Consequently superPoly agreement on smooth compactly supported tests in the common localization region implies superPoly agreement for every smooth polynomial-growth observable (`superPoly_moment_of_ccData`), and composing with the previous tide’s base-case-free headline yields `smooth_positive_jet_recovery_of_ccData` and `analytic_germ_recovery_of_ccData`: germbij Theorem 3.1 and Corollary 3.2, inverse directions, from the note’s own kind of data. With this, all three review findings are closed.

1 Setting

The fidelity review’s finding F3 was the quiet one: not a wrong statement, but a missing lemma. The note’s Φ_L is defined on $C_c^\infty(U)$; the Lean data lived on bare monomials against the U -restricted Gibbs measure. The gap “the difference is exponentially small by the loss gap” was asserted in the review and believed by everyone — and was exactly the kind of believed-but-unformalised step this perimeter arc exists to eliminate. The deliberation settled three design points: the tail estimate splits the rescaled exponent as $G \geq \frac{1}{2}c\|x\|^2 + \frac{1}{2}cr^2/q^2$ (both lower bounds hold on the tail support, so their average does); linearity happens at the *rescaled* level, where the merged integrability certificates already live; and the data premise quantifies over tests supported in the *common* localization region of the two packages — faithful to the note (which has one U) without pretending the two Lean packages share theirs.

2 The thing formalised

Three layers. First, a small missing vocabulary item: an eventual bound $|f(t)| \leq Ke^{-\delta t}$ with $\delta > 0$ makes f superpolynomially small (`superPoly_of_eventually_abs_le_exp`) — the quantitative form of the note’s “exponentially small terms lie below every element of the scale”. Second, the tail lemma (`posteriorMoment_cutoff_tail`): for continuous polynomial-growth P and continuous $0 \leq \chi \leq 1$ with $\chi \equiv 1$ on a ball of radius r around the minimum,

$$t \mapsto \langle P \rangle_t - \langle P\chi \rangle_t \text{ is smaller than every power of } t^{-1}.$$

¹2026-08-10-01-29-review-germbij-new-content; predecessors: the $k = 2$ bridge and the base-case discharge.

Third, the bridge and the composed corollaries: if the moment families of two localized losses agree beyond all orders on every smooth compactly supported test in the common localization region, then they agree on every smooth polynomial-growth observable; feeding the second-moment and monomial instances into the base-case-free headline gives positive-order jet equality and, for analytic losses, germ equality modulo the additive constant — the note-literal inverse of Theorem 3.1, with no base-case assumptions, no shared package matrix, and the note’s own observable class.

3 Proof strategy

The tail estimate works at the rescaled level, where the seabed keeps its integrability. At scale q the moment difference is a quotient of rescaled integrals with the dilation prefactors cancelled exactly (`posteriorMoment_eq_integrand_div`). The numerator’s integrand is supported where $q \cdot x \in U$ and $\|q \cdot x\| \geq r$ (inside the bump’s plateau it vanishes); there the coercivity field gives both $G \geq c\|x\|^2$ and $G \geq cr^2/q^2$ for the rescaled exponent G , so

$$e^{-G} \leq e^{-\frac{cr^2}{2q^2}} e^{-\frac{c}{2}\|x\|^2},$$

and with $|P(q \cdot x)| \leq C(1 + \|x\|^n)$ for $q \leq 1$ the whole numerator is dominated by $CMe^{-\frac{cr^2}{2q^2}}$ with M a fixed polynomial-Gaussian integral. The denominator tends to the positive Gaussian mass (the merged `tendsto_integral_rescaled_one`), so it is eventually at least half of it, and the quotient bound survives the $t = q^{-2}$ substitution as $Ke^{-\delta t}$. The bridge picks a `ContDiffBump` with plateau $\rho/2$ and support $3\rho/4$ inside both packages’ balls (ρ the smaller of the two radii), splits $m_A(P) - m_B(P)$ into two tails and one data term at $\varphi = P\chi$ (smooth times smooth, compactly supported, supported in both U ’s), and closes with superPoly additivity. The corollaries instantiate at $w_i w_j$ and `monomialTest` — coordinate projections are continuous linear maps, so smoothness is `contDiff_prod` over them.

4 Roadblocks and resolutions

All Lean-mechanical, in three families. The catalogued un-beta-reduced-lambda class struck twice more (after `integral_const_mul` and inside the indicator case split); `beta_reduce` and `show` dispatch it. `set-bound` locals are opaque to `positivity` (the bump constructor’s side conditions needed `linarith` against the recorded positivity fact instead). And two elaboration-order items: the exponential constant K of the superPoly source must be passed explicitly at `refine` time (it is not determined until the eventual bound is supplied), and a chained `closedBall ⊆ ball ⊆ ball` inclusion left its middle radius unsynthesizable — one direct `closedBall_subset_ball` step with the end radius was both simpler and elaborable. Four build iterations to green.

5 What was Mathlib, what was new

Mathlib lookup: the `ContDiffBump` API (plateau, support, compact support, smoothness; the inner-product-space existence instance), `tendsto_pow_mul_exp_neg_atTop_nhds_zero`, `norm_integral_le_of_norm_le`, `contDiff_prod`, `EuclideanSpace.proj`, `tsupport/support` plumbing. Seabed reuse: the coercivity field at $q = 1$, the dilation identity, the rescaled integrability certificate, the rescaled partition limit, and the previous tides’ freed headline. New: the exponential-to-superPoly packaging, the tail estimate, the cutoff bridge, and the note-literal

corollaries. The superPoly `add/sub` helpers and `HasPolynomialGrowth.comp_smul` are small general-purpose additions the seabed had been implicitly re-deriving.

6 Lessons

The review-to-tide pipeline closed a full loop here: a fidelity finding of the form “this implication is believed but not formalised” converts directly into a tide whose statement is the finding. Worth keeping: the finding’s own phrasing (“the difference is $O(e^{-\delta t})$ by the loss gap, hence superPoly-invisible”) was already the correct proof outline, and the deliberation only had to pin the split constant and the linearity altitude. Also catalogued for `CLAUDE.md`: `positivity` does not see through `set`, and implicit constants of eventual-bound lemmas need explicit instantiation under `refine`.

7 Follow-ups

- The located first-moment analogue: compose location recovery with the CC-data premise (the deliberation’s candidate C), giving the full note-literal Theorem 3.1 package — location, jet, and germ from one data premise.
- A single “grand corollary” bundling location + jet + germ, worth considering once the located CC form exists.
- Deriving `IsSymm` at order 2 from `ContDiff` remains the last hypothesis-list debt of the freed headlines.
- The tail lemma’s $0 \leq \chi \leq 1$ hypotheses could weaken to boundedness; not needed by any current caller.